

MATRIX CHAIN MULTIPLICATION (MCM)

1. Introduction

Matrix Chain Multiplication (MCM) is a Dynamic Programming problem whose objective is:

- To find the order of multiplying matrices that minimizes the total number of scalar multiplications.
- MCM does not perform the multiplication of matrices.

It only finds:

- Minimum multiplication cost.
- Best parenthesization (order of multiplication).

2. Why is MCM Needed?

Suppose:

- A_1 (30×15)
- A_2 (15×5)
- A_3 (5×10)
- A_4 (10×20)

Different orders of multiplication produce the same final matrix but require different numbers of scalar multiplications. Therefore, we must determine the order that gives minimum cost.

3. Matrix Dimension Array Representation

Given:

- $A_1 = 30 \times 15$
- $A_2 = 15 \times 5$
- $A_3 = 5 \times 10$
- $A_4 = 10 \times 20$

Dimension array:

$$P = [30, 15, 5, 10, 20]$$

where:

- $A_i = P_0 \times P_i$

- $A_2 = P_1 \times P_2$

- $A_3 = P_2 \times P_3$

- $A_4 = P_3 \times P_4$

4. State Definition

The state cell is defined as:

$$M[i, j] = \text{Minimum cost of multiplying } A_i A_{i+1} \dots A_j$$

5. Base Case

If only one matrix exists,

$$M[i, i] = 0$$

because no multiplication is needed. Therefore:

- $M[1, 1] = 0$

- $M[2, 2] = 0$

- $M[3, 3] = 0$

- $M[4, 4] = 0$

Cost Matrix after Base Case

$i \setminus j$	1	2	3	4
1	0	-	-	-
2	-	0	-	-
3	-	-	0	-
4	-	-	-	0

6. Recurrence Formula

For every possible split k such that $i \leq k < j$:

$$M[i, j] = \min_{i \leq k < j} \{ M[i, k] + M[k+1, j] + P_{i-1} P_k P_j \}$$

where:

- $M[i, k]$ = cost of left subchain
- $M[k+1, j]$ = cost of right subchain
- $P_{i-1}P_kP_j$ = cost of multiplying the two resulting matrices

7. Meaning of k

k is NOT the number of splits. k indicates the matrix where the left part ends.

General form:

$$(A_i \dots A_k)(A_{k+1} \dots A_j)$$

Therefore:

- Left part $\rightarrow A_i \dots A_k$
- Right part $\rightarrow A_{k+1} \dots A_j$

Example: $A_1A_2A_3A_4$

Possible splits:

- $k = 1 \rightarrow (A_1)(A_2A_3A_4)$
- $k = 2 \rightarrow (A_1A_2)(A_3A_4)$
- $k = 3 \rightarrow (A_1A_2A_3)(A_4)$

8. Initial Cost Matrix M

i \ j	1	2	3	4
1	0	-	-	-
2	-	0	-	-
3	-	-	0	-
4	-	-	-	0

9. Chain Length = 2

M[1, 2]

Only one split is possible: $k = 1$

Formula:

$$M[1, 2] = M[1, 1] + M[2, 2] + P_0P_1P_2$$

Substituting values: $M[1, 1] = 0$, $M[2, 2] = 0$, $P_0 = 30$, $P_1 = 15$, $P_2 = 5$

$$M[1, 2] = 0 + 0 + (30 \times 15 \times 5) = 2250$$

Hence: $M[1, 2] = 2250$, $K[1, 2] = 1$

M[2, 3]

Formula:

$$M[2, 3] = M[2, 2] + M[3, 3] + P_1P_2P_3$$

Substituting values: $M[2, 2] = 0$, $M[3, 3] = 0$, $P_1 = 15$, $P_2 = 5$, $P_3 = 10$

$$M[2, 3] = 0 + 0 + (15 \times 5 \times 10) = 750$$

Hence: $M[2, 3] = 750$, $K[2, 3] = 2$

M[3, 4]

Formula:

$$M[3, 4] = M[3, 3] + M[4, 4] + P_2P_3P_4$$

Substituting values: $M[3, 3] = 0$, $M[4, 4] = 0$, $P_2 = 5$, $P_3 = 10$, $P_4 = 20$

$$M[3, 4] = 0 + 0 + (5 \times 10 \times 20) = 1000$$

Hence: $M[3, 4] = 1000$, $K[3, 4] = 3$

Cost Matrix after Chain Length = 2

i \ j	1	2	3	4
1	0	2250	-	-
2	-	0	750	-
3	-	-	0	1000
4	-	-	-	0

10. Chain Length = 3

M[1, 3]

Two possible splits exist.

Split k = 1: $(A_1)(A_2A_3)$

$$M[1, 3] = M[1, 1] + M[2, 3] + P_0P_1P_3$$

Substituting values: $M[1, 1] = 0$, $M[2, 3] = 750$, $P_0 = 30$, $P_1 = 15$, $P_3 = 10$

$$M[1, 3] = 0 + 750 + (30 \times 15 \times 10) = 0 + 750 + 4500 = 5250$$

Split k = 2: $(A_1A_2)(A_3)$

$$M[1, 3] = M[1, 2] + M[3, 3] + P_0P_2P_3$$

Substituting values: $M[1, 2] = 2250$, $M[3, 3] = 0$, $P_0 = 30$, $P_2 = 5$, $P_3 = 10$

$$M[1, 3] = 2250 + 0 + (30 \times 5 \times 10) = 2250 + 1500 = 3750$$

Minimum cost = 3750. Hence: $M[1, 3] = 3750$, $K[1, 3] = 2$

M[2, 4]

Two possible splits exist.

Split k = 2: $(A_2)(A_3A_4)$

$$M[2, 4] = M[2, 2] + M[3, 4] + P_1P_2P_4$$

Substituting values: $M[2, 2] = 0$, $M[3, 4] = 1000$, $P_1 = 15$, $P_2 = 5$, $P_4 = 20$

$$M[2, 4] = 0 + 1000 + (15 \times 5 \times 20) = 0 + 1000 + 1500 = 2500$$

Split k = 3: $(A_2A_3)(A_4)$

$$M[2, 4] = M[2, 3] + M[4, 4] + P_1P_3P_4$$

Substituting values: $M[2, 3] = 750$, $M[4, 4] = 0$, $P_1 = 15$, $P_3 = 10$, $P_4 = 20$

$$M[2, 4] = 750 + 0 + (15 \times 10 \times 20) = 750 + 3000 = 3750$$

Minimum cost = 2500. Hence: $M[2, 4] = 2500$, $K[2, 4] = 2$

Cost Matrix after Chain Length = 3

i \ j	1	2	3	4
1	0	2250	3750	-
2	-	0	750	2500
3	-	-	0	1000
4	-	-	-	0

11. Chain Length = 4

M[1, 4]

Three possible splits exist.

Split k = 1: (A₁)(A₂A₃A₄)

$$M[1, 4] = M[1, 1] + M[2, 4] + P_0P_1P_4$$

Substituting values: $M[1, 1] = 0$, $M[2, 4] = 2500$, $P_0 = 30$, $P_1 = 15$, $P_4 = 20$

$$M[1, 4] = 0 + 2500 + (30 \times 15 \times 20) = 0 + 2500 + 9000 = 11500$$

Split k = 2: (A₁A₂)(A₃A₄)

$$M[1, 4] = M[1, 2] + M[3, 4] + P_0P_2P_4$$

Substituting values: $M[1, 2] = 2250$, $M[3, 4] = 1000$, $P_0 = 30$, $P_2 = 5$, $P_4 = 20$

$$M[1, 4] = 2250 + 1000 + (30 \times 5 \times 20) = 2250 + 1000 + 3000 = 6250$$

Split k = 3: (A₁A₂A₃)(A₄)

$$M[1, 4] = M[1, 3] + M[4, 4] + P_0P_3P_4$$

Substituting values: $M[1, 3] = 3750$, $M[4, 4] = 0$, $P_0 = 30$, $P_3 = 10$, $P_4 = 20$

$$M[1, 4] = 3750 + 0 + (30 \times 10 \times 20) = 3750 + 6000 = 9750$$

Minimum cost = 6250. Hence: $M[1, 4] = 6250$, $K[1, 4] = 2$

Cost Matrix after Chain Length = 4

i \ j	1	2	3	4
1	0	2250	3750	6250
2	-	0	750	2500
3	-	-	0	1000
4	-	-	-	0

12. Final Cost Matrix M

$i \setminus j$	1	2	3	4
1	0	2250	3750	6250
2	-	0	750	2500
3	-	-	0	1000
4	-	-	-	0

13. Final K Matrix

$i \setminus j$	1	2	3	4
1	-	1	2	2
2	-	-	2	2
3	-	-	-	3
4	-	-	-	-

14. Why Do K Values Change?

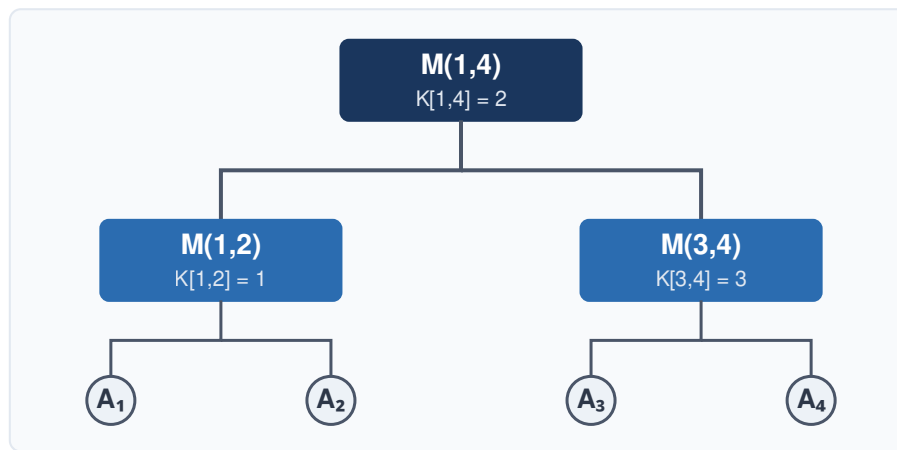
K stores the split position that gives minimum cost. Examples:

- For $M[1, 3]$: $k=1 \rightarrow 5250$, $k=2 \rightarrow 3750$. Since 3750 is smaller, $K[1, 3] = 2$.
- For $M[1, 4]$: $k=1 \rightarrow 11500$, $k=2 \rightarrow 6250$, $k=3 \rightarrow 9750$. Minimum occurs at $k=2$, so $K[1, 4] = 2$.

Thus, K values are not random. They depend upon which split produces minimum cost.

15. Box Flow Diagram

The step-by-step reconstruction of the optimal split sequence using the K Matrix values:



16. Parenthesization

Reconstructed hierarchy flow:

- $M(1,4) \downarrow$ Split at $k=2$
- $(M_{1,2})(M_{3,4})$
- $(A_1A_2)(A_3A_4)$

Optimal Parenthesization: $((A_1A_2)(A_3A_4))$

17. Final Answers

- (a) **Minimum Scalar Multiplications:** 6250
- (b) **Optimal Parenthesization:** $((A_1A_2)(A_3A_4))$
- (c) **Cost Matrix M & K Matrix:** (Refer to Section 12 and Section 13 above for complete values)

18. Time Complexity

- Number of states = $O(n^2)$
- For every state we try $O(n)$ values of k
- **Time Complexity:** $O(n^3)$
- **Space Complexity:** $O(n^2)$

19. Last Minute Revision

- **State:** $M[i, j]$ = Minimum cost of multiplying matrices i to j .
- **Base Case:** $M[i, i] = 0$
- **Formula:** $M[i, j] = \min \{ M[i, k] + M[k+1, j] + P_{i-1}P_kP_j \}$

- **K Matrix:** Stores the split position that gives minimum cost. It is not random.
- **Parenthesization:** Reconstructed using the K matrix.
- **Complexities:** Time = $O(n^3)$ | Space = $O(n^2)$